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Food system drivers of global foodborne bacterial disease burden: Pathogen-specific attribution using 1990–2021 GBD 2021 and FAO food system databases, with implications for food safety control

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ABSTRACT

Foodborne bacterial pathogens cause significant global illness and death, but data on their food transmission and links to food systems are limited. This study integrates the Global Burden of Disease (GBD) 2021, WHO-derived food-attribution fractions (FAFs), and the FAO food system across 204 countries (1990–2021). It quantifies the foodborne-attributable burden of bacterial pathogens (*EPEC*, *ETEC*, *non-typhoidal Salmonella*, *Shigella*, *Campylobacter*, and *L. monocytogenes*) and examines associations within the food system through network pathway analysis. These pathogens caused an estimated 364,500 deaths from diarrheal diseases (age-standardized death rate, ASDR 3.63/100,000) and 11,100 from meningitis (ASDR 0.12/100,000), with corresponding DALYs of 24.61 million (age-standardized DALY rate, ASDALY 312.95/100,000) and 699,500 (ASDALY 8.90/100,000) in 2021. The Burden was concentrated in Sub-Saharan Africa and South Asia (78.7% of global DALYs), with low-SDI regions experiencing 195-fold higher diarrheal diseases ASDALY than high-SDI settings. Age-standardized rates declined substantially (estimated annual percentage change, EAPC of diarrheal diseases: −4.35%), yet high-SDI countries exhibited stagnation in diarrheal diseases mortality (ASDR EAPC: +0.63%). Network pathway analysis revealed that socioeconomic development (SDI: $\rho = -0.830$) mediated unexpected production-burden relationships. Animal-source food production negatively correlated with pathogen burden ($\rho = -0.28$ to -0.22), indicating food safety capacity, while plant-associated pathogens positively correlated with vegetable and legume production ($\rho = +0.20$ to $+0.41$) in low-SDI settings. Projections indicate declines in meningitis (64.59%) and diarrheal diseases (69.50%) by 2050, with Sub-Saharan Africa facing a burden 51.66 times higher for diarrheal diseases and 29.73 times higher for meningitis than high-income regions. These findings indicate food safety infrastructure, not production volume, influences foodborne disease outcomes, supporting targeted interventions based on pathogen–food combinations and regional contexts.

1. Introduction

As a persistent public health challenge, foodborne diseases triggered by bacterial pathogens cause high morbidity and mortality globally. These pathogens are transmitted predominantly through contaminated food, are capable of causing severe human disease, particularly in vulnerable populations including pregnant women, the elderly, and children, and are widely distributed throughout food production and supply chains (Hoffmann et al., 2017; World Health Organization, 2015). The World Health Organization Foodborne Disease Burden Epidemiology Reference Group reported approximately 420,000 deaths and 33 million disability-adjusted life years (DALYs) annually attributable to foodborne pathogen infection (Afsana et al., 2022; Havelaar et al., 2015).

Since 1990, global food systems have undergone profound transformations due to the population growth, urbanization, economic development, and dietary transitions toward increased consumption of animal-source foods (Godfray et al., 2018). It has been reported that

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global meat production has increased more than threefold over the past 50 years with annual yields now surpassing 350 million tonnes. The most dramatic expansion has been observed in low and middle-income countries (LMICs), a trend underpinned by the ongoing growth in demand for animal-source foods (Ritchie et al., 2019). Additionally, international food trade has expanded substantially, likely leading to more interconnected pathogen transmission pathways of foodborne pathogens among countries (Food and Agriculture Organization, 2022). Driven by this sweeping transformation, LMICs are confronting acute food safety challenges, stemming from the disconnect between rapid agricultural intensification and the slow development of food safety infrastructure such as cold chain systems, veterinary inspection services, and microbiological monitoring capacity. The World Bank has emphasized the economic imperative for strengthening food safety systems in LMICs (Jaffee et al., 2018). While high-income countries have advanced food safety regulatory frameworks, they also face emergent challenges, owing to the rising antimicrobial resistance among foodborne pathogens, consumer preferences for minimally processed and ready-to-eat foods, and vulnerabilities associated with extended supply chains (Newell et al., 2010; Tang et al., 2017). Thus, accurate, pathogen-specific burden estimates are required to effectively inform intervention prioritization and provide useful information, especially for nations with limited food safety resources.

Current assessments of foodborne disease burden face methodological challenges that limit their utility for targeted intervention design. The Global Burden of Disease (GBD) Study provides comprehensive burden estimates for 204 countries and territories from 1990 to 2021, quantifying incidence, mortality, and DALYs for 371 diseases and injuries through synthesis of over 100,000 data sources (Ferrari et al., 2024; Naghavi et al., 2024). However, GBD estimates for bacterial pathogens encompass all transmission routes (waterborne, person-to-person, foodborne, and environmental) without pathway-specific attribution, thereby obscuring the specific contribution of contaminated food to disease burden, and interpreting total pathogen burden as foodborne-specific burden introduces substantial overestimation. To address this, WHO-FERG has developed standardized methodologies for estimating food attribution fractions (FAFs) that partition multi-route pathogen burden into transmission-specific components through structured expert elicitation integrating systematic literature reviews, outbreak surveillance data, and region-specific expert judgment (Devleesschauwer et al., 2015; Hald et al., 2016; Hoffmann et al., 2017). To date, validated region-specific FAFs have been established for a defined subset of bacterial pathogens, including *non-typhoidal Salmonella (NTS)* and *Campylobacter*, the leading causes of gastroenteritis linked to contaminated meat and poultry (Ehuwa et al., 2021; Zizza et al., 2024); enteropathogenic and enterotoxigenic *Escherichia coli (EPEC and ETEC)*, which drive acute diarrheal illness predominantly in low-hygiene settings (Croxen et al., 2013); *Shigella*, the agent of bacillary dysentery with regionally variable transmission routes (Hald et al., 2016); and *Listeria monocytogenes (L. monocytogenes)*, which poses disproportionate risks to immunocompromised and perinatal populations via ready-to-eat foods (Yangchen et al., 2025). The availability of validated FAFs for all six pathogens enables pathway-specific burden quantification and accordingly defines the pathogen scope of this study. Nevertheless, the extent to which these attribution estimates have been systematically integrated with longitudinal GBD estimates and food system indicators remains limited, and the dynamic associations between evolving food systems and foodborne disease trajectories over extended time horizons warrant further investigation.

Thus, the objective of this study is to develop a methodological framework by integrating GBD 2021 data, pathogen and region-specific food attribution fractions derived from WHO structured expert elicitation studies, and food system indicators from FAO databases, enabling comprehensive food system analyses and providing evidence for food safety policy prioritization. Firstly, FAO food system databases encompassing production, trade, and consumption indicators across major food commodity categories were incorporated to conduct comprehensive correlation and temporal trend analyses. Subsequently, correlation-based network analysis integrated with graph neural network (GNN) methods was employed to construct multidimensional pathway models linking socioeconomic development, food production systems, and pathogen-specific disease outcomes, capturing both linear associations and higher-order structural relationships among variables. Finally, ensemble time-series projection models were developed to predict disease burden trajectories by 2050. These findings provide pathogen- and region-specific evidence for informing targeted food safety interventions and policy decisions aimed at reducing the burden of foodborne diseases.

2. Methods

2.1. Data sources and statistical analysis

2.1.1. Pathogen selection and GBD data

This study focuses on six major bacterial pathogens (*EPEC*, *ETEC*, *NTS*, *Shigella*, *Campylobacter*, and *L. monocytogenes*) associated with foodborne transmission. These pathogens collectively account for a substantial share of the global foodborne disease burden (Havelaar et al., 2015; Kirk et al., 2015; Tibebu et al., 2024). According to the Global Burden of Disease (GBD) database (https://www.healthdata.org/research-analysis/gbd), the main disease categories linked to these pathogens are diarrheal diseases and meningitis, which represent the primary clinical outcomes of foodborne exposure (GBD 2021 Diarrhoeal Diseases Collaborators, 2025; Global Burden of Disease Collaborative Network, 2024).

For each pathogen-disease combination, annual estimates of incidence, deaths, and DALYs with 95% uncertainty intervals (UIs) were extracted for 204 countries and territories from 1990 to 2021. Extracted data encompassed all-age counts, age-standardized rates, and age-sex-stratified estimates, with the latter further disaggregated by country, year, sex, and five aggregated age groups. Geographic stratification followed GBD multi-level regional standards to enable comprehensive spatial analysis. Supplementary datasets from the FAO, World Bank, and WHO, covering agricultural production, food trade flows, consumption patterns, and economic indicators, were additionally integrated and harmonized to the GBD seven super-region framework to ensure cross-source consistency.

2.1.2. Food attribution fractions

To derive foodborne-specific burden estimates from GBD total burden data that encompass multiple transmission pathways, pathogen- and disease-specific food attribution fractions (FAFs) were applied. The region-specific FAFs were obtained from, multiple sources depending on pathogen characteristics (Table 1). Specifically, *EPEC*, *ETEC*, *NTS*, *Shigella*, and *Campylobacter* were retrieved from a WHO structured expert elicitation study that quantified the relative contributions of foodborne transmission to the overall disease burden associated with these priority pathogens across global super-regions (Hald et al., 2016). In the case of *L. monocytogenes*, a fixed foodborne proportion of 1.00 (1.00–1.00) was assigned to all super-regions, which is corroborated by WHO estimates indicating the well-established predominance of foodborne transmission for this pathogen (Hald et al., 2016).

The foodborne-attributable burden for the stratification units, including location, sex, age, disease, pathogen, and year, was calculated by Eq. (1), where pathogen-specific foodborne proportions corresponding to the super-region of each location were multiplied by the original GBD all-route transmission burden. The same proportions were used in both point estimates with the corresponding lower and upper uncertainty bounds to maintain consistency. After pathogen-specific attribution, the aggregated foodborne burden was computed by summing the foodborne-attributable burden across the six pathogens within each stratification unit using Eq. (2). The pathogen-specific estimates

Table 1
Foodborne proportion of priority pathogens across global super-regions.

Super-region	Food attribution fraction, mean (95% UI)					
	EPEC	***ETEC***	***NTS***	***Shigella***	***Campylobacter***	***L. monocytogenes***
1. Sub-Saharan Africa	0.29 (0.02–0.62)	0.33 (0.08–0.65)	0.46 (0.12–0.74)	0.15 (0.00–0.52)	0.57 (0.30–0.75)	1.00 (1.00–1.00)
2. High-income	0.68 (0.18–0.94)	0.39 (0.10–0.69)	0.74 (0.43–0.93)	0.11 (0.00–0.47)	0.72 (0.41–0.91)	1.00 (1.00–1.00)
3. Latin America and Caribbean	0.30 (0.02–0.62)	0.35 (0.08–0.67)	0.50 (0.12–0.75)	0.14 (0.00–0.52)	0.68 (0.39–0.82)	1.00 (1.00–1.00)
4. North Africa and Middle East	0.31 (0.06–0.62)	0.35 (0.06–0.66)	0.50 (0.19–0.75)	0.14 (0.00–0.52)	0.67 (0.40–0.82)	1.00 (1.00–1.00)
5. South Asia	0.29 (0.01–0.65)	0.38 (0.03–0.73)	0.56 (0.12–0.85)	0.35 (0.01–0.69)	0.54 (0.15–0.80)	1.00 (1.00–1.00)
6. Southeast Asia, East Asia, and Oceania	0.30 (0.01–0.62)	0.38 (0.03–0.72)	0.57 (0.25–0.82)	0.36 (0.01–0.70)	0.57 (0.25–0.82)	1.00 (1.00–1.00)
7. Central Europe, Eastern Europe, and Central Asia	0.48 (0.06–0.81)	0.43 (0.06–0.73)	0.62 (0.32–0.84)	0.15 (0.00–0.53)	0.66 (0.34–0.87)	1.00 (1.00–1.00)

Note: For pathogens with region-specific FAFs, values from (Hald et al., 2016) were applied according to super-region classification. For *L. monocytogenes*, uniform FAFs were applied across all super-regions based on predominant foodborne transmission.

were traced while providing the comprehensive aggregate measures of the total foodborne disease burden.

$$N_{food}(l,s,a,d,r,y) = p_{r,super(l)} \times N_{all}(l,s,a,d,r,y) \quad (1)$$

$$N^{all}_{food}(l,s,a,d,y) = \sum_{r} N_{food}(l,s,a,d,r,y) \quad (2)$$

where N_{food} denotes the foodborne-attributable burden (Deaths or DALYs); Nall represents the all-route transmission burden from GBD; $p_{r,\ super(l)}$ denotes the foodborne proportion for pathogen r in the super-region corresponding to location l; l denotes location; s denotes sex; a denotes age group; d denotes disease; r denotes pathogen; and y denotes year.

2.1.3. Age-specific and age-standardized rates

The age-specific mortality and DALY rates were calculated using Eq. (3). The age-standardized mortality rates (ASDR) and age-standardized DALYs rates (ASDALY) were then computed using Eq. (4) based on the direct standardization method with WHO World Standard Population age weights (Table S1) (Ahmad et al., 2001). Age-standardized rates (ASRs) were computed to enable valid cross-population comparisons, with the standardization procedure applied consistently to uncertainty bounds to ensure coherent uncertainty propagation throughout the analytical pipeline.

$$Age-Specific\ Rate = \frac{N_{food}}{P} \times 100,000 \quad (3)$$

$$ASR = \sum_{a} \left[\frac{w_a}{w} \bullet \left(\frac{N_{food}(a)}{P_a} \times 100,000 \right) \right] \quad (4)$$

where N_{food} denotes the foodborne-attributable burden (Deaths or DALYs) for a given stratification unit; P denotes the corresponding population size; $N_{food(a)}$ and P_a denote the age-specific foodborne-attributable burden and population size for age group a, respectively; w_a represents the WHO World Standard Population weight for age group a; w represents the total standard population weight; and ASR denotes age-standardized rate (ASDR or ASDALY).

2.2. Temporal trend analysis

To quantify the temporal trends in foodborne disease burden from 1990 to 2021, the estimated annual percentage change (EAPC) in age-standardized rates was calculated using log-linear regression. Specifically, the natural logarithm of age-standardized rates was regressed on calendar year using Eq. (5), and EAPC was derived from the regression coefficient β using Eq. (6). The 95% CI for EAPC was calculated using Eq. (7), applied consistently across all geographic hierarchies, demographic strata, and pathogen-specific estimates, following Hankey et al. and subsequent GBD-derived burden analyses (Hankey et al., 2000; Li et al., 2024).

$$\ln[ASR(t)] = \alpha + \beta t + \varepsilon \quad (5)$$

$$EAPC = 100 \times \left(e^{\beta} - 1\right) \quad (6)$$

$$95\%CI_{EAPC} = 100 \times \left[e^{\left(\beta \pm 1.96 \times SE_{\beta}\right)}\right. \quad (7)$$

where $ASR(t)$ denotes the age-standardized rate in year t; α and β denote the intercept and slope coefficients, respectively; ε denotes the error term; $EAPC$ stands for estimated annual percentage change; and SE_{β} denotes the standard error of β.

2.3. Food system and socioeconomic indicators

The food production and trade data were obtained from Food and Agriculture Organization Corporate Statistical Database (FAOSTAT) from 1990 to 2021. Primary agricultural and livestock products from FAO databases were divided into 8 major categories, including cereals and grains, egg products, fruits, legumes, meat products, milk and dairy products, nuts, and vegetables. Production values were measured in tonnes, and trade balance indicator values were calculated as net export values (exports minus imports, measured in 1000 USD) for each food category. Finally, several composite indicators were developed to capture multidimensional food system dynamics, including aspects of production efficiency, distribution equity, and consumption patterns, with detailed descriptions presented in Table S2. Specifically, the Food-to-General Price Ratio was obtained according to the ratio of food consumer price index to general consumer price index. Food Price Volatility was quantified according to three-year rolling standard deviation of food consumer prices. Food Affordability was computed according to GDP per capita divided by food consumer price index. Using the logarithmic transformations in GDP per capita and food affordability indicators normalizes their distributions. Food price indices (consumer food prices, general consumer prices) were derived from FAO food price monitoring databases. The urbanization rate was calculated as the ratio of the urban population to the total population.

For food price data available only from 2000 to 2021, historical values spanning 1990–1999 were imputed using a multi-step backward extrapolation approach. Regional economic projections were first generated via Compound Annual Growth Rate (CAGR) modeling with constrained annual bounds of −15% to +25% to avoid implausible volatility, followed by region-specific inflation adjustment to account for differential price dynamics across GBD super-regions, and finalized through three-period moving averages to ensure temporal stability at the transition interface between imputed and observed data. The methodology was validated against 2000–2010 data, accurately reconstructing known values across developed (High-income North America), transition (Eastern Europe), and developing regions (Central Sub-Saharan Africa), with regional correlation coefficients exceeding 0.90. Following successful validation, this framework was applied to impute values for 1990–1999, with regional growth rate distributions presented in Fig. S1.

2.4. Analytical framework for temporal projections and pathway analysis

This study employed a two-step analytical framework to characterize temporal trends and structural determinants of foodborne disease burden. The first step involved projecting future burden trajectories through ensemble forecasting, while the second step identified key determinants via correlation-based pathway analysis, augmented by graph neural network (GNN) methods to resolve higher-order structural relationships.

The ensemble forecasting approach combined four complementary methods to capture diverse temporal dynamics: autoregressive integrated moving average (ARIMA) models for autocorrelation structures; Prophet models for long-term trend decomposition; TBATS (Trigonometric seasonality, Box-Cox transformation, ARMA errors, Trend and Seasonal components) models for annual data patterns; and joinpoint regression to identify statistically significant trend inflection points. This methodology visualized potential burden trajectories, with all projections presented alongside 95% uncertainty intervals reflecting historical variance in burden estimates.

The rank of Spearman correlations examined associations between age-standardized burden rates and food system or socioeconomic indicators across country-year observations (1990–2021), with Benjamini-Hochberg FDR correction applied ($q < 0.05$). Significant associations were visualized, with nodes representing variables and edges weighted by absolute correlation coefficients ($|\rho|$). These associations were classified into four mechanistic categories: direct pathways connecting food system or socioeconomic variables to disease burden; intermediate pathways capturing inter-predictor correlations indicating mediated effects; pathogen-specific food vehicle pathways linking bacterial pathogens to established transmission vehicles based on epidemiological evidence; and regional distribution pathways capturing geographic clustering of pathogen burden across GBD super-regions. The variable importance was assessed by ranking absolute correlation coefficients derived from direct pathways, with variables classified as protective factors ($\rho < 0$) or risk factors ($\rho > 0$) based on the direction of association. Hub node centrality was quantified using degree centrality (number of significant connections) and weighted degree (sum of absolute coefficients) to identify potential mediators. Notably, the SDI was evaluated separately due to its theoretical role in coordinating food safety capacity.

To enhance traditional correlation-based network analysis, GNN learning was integrated to capture higher-order structural relationships among variables. Previous studies have demonstrated the effectiveness of GNNs in understanding complex relationships in various domains (Gkarmpounis et al., 2024). The GNN model utilized a three-layer graph convolutional network (GCN) architecture with graph attention mechanisms, learning node embeddings by propagating information through the correlation-based network structure. These learned embeddings were then used to re-weight existing edges identified by statistical correlation analysis, with edge weights adjusted according to cosine similarity between node embeddings, as shown in Eqs. (8) and (9):

$$weight_{ij} = Q_{ij} \times \left(1 + 0.5 \times \left|\cos\theta_{ij}\right|\right) \tag{8}$$

$$\cos\theta_{ij} = \frac{h_i \bullet h_j}{\|h_i\| \bullet \|h_j\|} \tag{9}$$

where Q_{ij} denotes the Spearman correlation coefficient between variables i and j; $\cos\theta_{ij}$ denotes the cosine similarity between GNN-learned node embeddings h_i and h_j. The scaling factor 0.5 constrains the maximum GNN-derived weight adjustment to 50% of the base correlation weight, ensuring that statistical associations identified by Spearman analysis remain the primary determinant of edge inclusion, while GNN structural similarity serves as a supplementary modulator. Edges were only retained if ρ_{ij} was statistically significant after FDR correction ($q < 0.05$), such that GNN similarity alone could not introduce edges absent from the correlation-based network.

The resulting integrated edge weights were used to construct a weighted network, in which nodes representing variables and edges weighted by the integrated correlation-GNN scores. To characterize the relative importance of individual variables within this network, nodes were ranked by the absolute magnitude of their direct Spearman correlation coefficients, with with the direction of association used to classifying variables as either protective ($\rho < 0$) or risk-associated ($\rho > 0$). Hub node centrality was additionally quantified to identify variables with disproportionate connectivity, providing a basis for inferring potential mediating roles within the network architecture.

2.5. Analysis software and computational environment

All statistical analyses were performed in Python 3.10, using pandas and NumPy for data processing; SciPy and statsmodels for statistical testing (Spearman correlation with FDR correction); pmdarima, Prophet, and TBATS for ARIMA modeling, trend decomposition, and time-series forecasting, respectively; PyTorch and PyTorch Geometric for graph neural network modeling.

3. Results and discussion

3.1. Global disease burden: magnitude, distribution, and temporal dynamics

Systematic attribution of foodborne bacterial pathogens to the global disease burden in 2021 revealed substantial yet inequitably distributed health impacts. As summarized in Table 2, diarrheal diseases accounted for 364,500 deaths (ASDR 3.63/100,000) and 24.61 million DALYs (ASDALY 312.95/100,000), while meningitis resulted in 11,100 deaths (ASDR 0.12/100,000) and 699,500 DALYs (ASDALY 8.90/100,000). Notably, the ASDR for foodborne diarrheal diseases alone (3.63/100,000) represents a substantial proportion of the WHO global all-cause foodborne benchmark (~5.3/100,000), and the combined burden of these two foodborne disease categories remains comparable in magnitude to major infectious diseases prioritized in LMICs (Grace, 2023). The disproportionate concentration of this burden in low- and middle-income settings, where per-capita food safety investment remains orders of magnitude lower than in high-income countries, suggests that observed burden differentials may predominantly reflect structural resource asymmetries rather than solely inherent pathogen-exposure differences. Given this substantial disease burden, comprehensive food safety governance frameworks spanning production, processing, and distribution systems constitute essential public health imperatives (Havelaar et al., 2015).

Pathogen–disease–region connectivity analysis through Sankey diagram (Fig. 1) revealed the proportional attributed burden flowing from six priority pathogens through two syndromes across 21 GBD regions. The analysis demonstrated distinct geographic stratification in pathogen dominance: *NTS*, *ETEC*, and *Shigella* dominated diarrheal disease burden in South Asia, Sub-Saharan Africa, and Southeast Asia, where they collectively accounted for >70% of diarrheal diseases burden; *Campylobacter* exhibited elevated proportions in high-income regions and was the primary pathogen in Western Europe; *EPEC* showed consistent representation across regions at 10–15% of diarrheal diseases burden; and *L. monocytogenes* exhibited a narrow yet distinct flow primarily toward meningitis, consistent with its invasive phenotype. *NTS* additionally displayed multi-syndrome pathogenicity with flow bands extending toward both diarrheal diseases and meningitis categories. The disease-to-region flows demonstrated that South Asia, Eastern and Western Sub-Saharan Africa, and Southeast Asia received the widest aggregate bands, whereas High-income regions received comparatively narrow bands. Although Water, Sanitation and Hygiene (WASH) infrastructure expansion has driven measurable improvements in LMICs, incomplete control of environmental fecal contamination sustains a

Table 2
Estimated burden of diseases associated with priority foodborne pathogens: mortality, disability-adjusted life years (DALYs), and temporal trends by region, sex, and pathogen (Global Burden of Disease Study 1990–2021).

	Mortality			DALYs		
	Cases in 2021 (10,000 persons,95% UI)	ASDR in 2021 (/100,000) (95% UI)	EAPC for 1990 to 2021, % (95% CI)	Cases in 2021 (10,000 persons, 95% UI)	ASDALY in 2021 (/100,000) (95% UI)	EAPC for 1990 to 2021, % (95% CI)
Global						
MEN	1.11 (0.82, 1.48)	0.12 (0.09, 0.16)	−2.37 (−2.57, −2.17)	69.95 (49.44, 95.53)	8.90 (6.41, 11.88)	−2.46 (−2.69, −2.23)
DIA	36.45 (16.23, 71.09)	3.63 (1.81, 6.54)	−4.44 (−4.75, −4.13)	2460.78 (1236.40, 4450.24)	312.95 (164.59, 542.73)	−4.35 (−4.67, −4.03)
Gender						
1 Male						
MEN	0.59 (0.43, 0.82)	0.13 (0.09, 0.18)	−2.41 (−2.61, −2.22)	38.18 (26.28, 54.62)	9.54 (6.70, 13.34)	−2.49 (−2.71, −2.28)
DIA	18.53 (7.88, 37.39)	3.81 (1.80, 7.16)	−4.41 (−4.68, −4.14)	1312.68 (630.83, 2458.78)	327.95 (164.78, 590.54)	−4.32 (−4.59, −4.05)
2 Female						
MEN	0.52 (0.38, 0.70)	0.11 (0.08, 0.15)	−2.32 (−2.53, −2.10)	31.77 (22.30, 44.45)	8.22 (5.87, 11.29)	−2.41 (−2.66, −2.17)
DIA	17.92 (7.06, 37.50)	3.44 (1.62, 6.37)	−4.48 (−4.84, −4.12)	1148.10 (549.81, 2128.12)	296.98 (151.16, 521.22)	−4.38 (−4.75, −4.00)
SDI rank						
1 High SDI						
MEN	0.03 (0.02, 0.03)	0.01 (0.01, 0.01)	−3.65 (−3.85, −3.46)	0.89 (0.73, 1.07)	0.73 (0.64, 0.83)	−3.97 (−4.16, −3.79)
DIA	0.21 (0.05, 0.48)	0.02 (0.01, 0.05)	0.63 (0.13, 1.12)	5.37 (2.05, 10.72)	3.78 (1.87, 6.47)	−0.13 (−0.56, 0.30)
2 High-middle SDI						
MEN	0.08 (0.06, 0.10)	0.02 (0.02, 0.03)	−5.56 (−5.82, −5.30)	3.30 (2.56, 4.17)	1.52 (1.27, 1.80)	−6.18 (−6.42, −5.95)
DIA	0.18 (0.06, 0.39)	0.06 (0.03, 0.11)	−10.92 (−11.20, −10.65)	11.62 (5.20, 22.21)	6.70 (3.44, 11.48)	−10.34 (−10.60, −10.08)
3 Middle SDI						
MEN	0.07 (0.05, 0.10)	0.05 (0.04, 0.06)	−3.00 (−3.17, −2.82)	3.75 (2.68, 5.13)	3.05 (2.47, 3.75)	−3.46 (−3.64, −3.29)
DIA	1.13 (0.43, 2.40)	0.75 (0.35, 1.35)	−5.87 (−6.00, −5.74)	79.72 (35.72, 154.30)	67.27 (34.89, 114.87)	−5.83 (−5.96, −5.70)
4 Low-middle SDI						
MEN	0.56 (0.36, 0.85)	0.18 (0.13, 0.24)	−2.33 (−2.56, −2.11)	37.63 (23.46, 58.40)	12.68 (8.92, 17.57)	−2.30 (−2.56, −2.05)
DIA	23.97 (9.39, 50.53)	5.75 (2.75, 10.30)	−5.06 (−5.32, −4.79)	1487.46 (673.58, 2862.07)	473.06 (245.69, 802.67)	−4.89 (−5.16, −4.62)
5 Low SDI						
MEN	0.35 (0.21, 0.55)	0.41 (0.31, 0.54)	−2.96 (−3.13, −2.78)	22.97 (13.92, 37.18)	25.49 (18.96, 34.47)	−3.23 (−3.40, −3.06)
DIA	10.04 (4.16, 20.56)	9.18 (4.42, 16.39)	−4.65 (−4.84, −4.45)	808.23 (353.83, 1612.20)	736.99 (381.61, 1270.86)	−4.71 (−4.91, −4.51)
GBD regions						
1 Central Europe, Eastern Europe, and Central Asia						
MEN	0.02 (0.02, 0.03)	0.03 (0.03, 0.04)	−4.02 (−4.36, −3.68)	0.85 (0.65, 1.08)	1.99 (1.76, 2.29)	−4.46 (−4.76, −4.15)
DIA	0.03 (0.00, 0.11)	0.07 (0.01, 0.23)	−6.64 (−7.08, −6.19)	2.20 (0.23, 7.55)	7.01 (0.86, 21.12)	−6.45 (−6.86, −6.05)
2 High income						
MEN	0.03 (0.03, 0.04)	0.01 (0.01, 0.01)	−3.54 (−3.71, −3.37)	0.98 (0.81, 1.17)	0.80 (0.70, 0.91)	−3.92 (−4.07, −3.76)
DIA	0.13 (0.01, 0.42)	0.02 (0.00, 0.04)	−1.51 (−1.82, −1.20)	3.31 (0.53, 9.32)	2.22 (0.47, 5.46)	−1.51 (−1.87, −1.15)
3 Latin America and Caribbean						
MEN	0.03 (0.02, 0.04)	0.04 (0.03, 0.05)	−4.24 (−4.47, −4.01)	1.58 (1.16, 2.12)	2.65 (2.19, 3.22)	−4.65 (−4.86, −4.44)
DIA	0.16 (0.02, 0.57)	0.20 (0.03, 0.63)	−7.63 (−7.95, −7.30)	10.03 (1.34, 34.39)	17.16 (2.63, 52.98)	−7.63 (−7.92, −7.33)
4 North Africa and Middle East						
MEN	0.03 (0.02, 0.05)	0.04 (0.03, 0.06)	−3.92 (−4.20, −3.64)	1.99 (1.27, 3.07)	2.90 (2.15, 3.93)	−4.15 (−4.48, −3.83)
DIA	0.15 (0.02, 0.60)	0.20 (0.03, 0.69)	−6.97 (−7.21, −6.74)	13.37 (1.70, 51.02)	19.32 (2.89, 64.37)	−6.69 (−6.91, −6.48)
5 South Asia						
MEN	0.24 (0.17, 0.34)	0.12 (0.09, 0.14)	−3.00 (−3.21, −2.78)	15.06 (10.12, 21.61)	7.99 (6.04, 10.23)	−2.94 (−3.20, −2.68)
DIA	5.49 (0.20, 21.83)	1.46 (0.06, 4.98)	−6.45 (−6.70, −6.20)	238.75 (11.76, 864.12)	109.69 (6.34, 345.19)	−6.39 (−6.64, −6.14)
6 Southeast Asia, East Asia, and Oceania						
MEN	0.11 (0.08, 0.15)	0.04 (0.03, 0.06)	−4.18 (−4.31, −4.04)	6.21 (4.34, 8.62)	3.28 (2.44, 4.27)	−4.44 (−4.60, −4.28)
DIA	0.72 (0.03, 2.69)	0.21 (0.01, 0.69)	−7.48 (−7.62, −7.34)	36.25 (2.32, 124.28)	19.00 (1.42, 58.62)	−7.34 (−7.49, −7.20)
7 Sub-Saharan Africa						
MEN	0.62 (0.38, 0.99)	0.47 (0.34, 0.65)	−2.53 (−2.74, −2.33)	41.87 (25.01, 68.27)	29.57 (20.99, 41.57)	−2.74 (−2.95, −2.53)
DIA	5.70 (0.42, 24.36)	3.36 (0.27, 12.81)	−4.33 (−4.67, −3.99)	461.61 (36.33, 1941.31)	271.71 (24.64, 1008.04)	−4.34 (−4.69, −3.99)

Notes: MEN, meningitis; DIA, diarrheal diseases. ASDR/ASDALY, age-standardized mortality/DALYs rate per 100,000 population. EAPC, estimated annual percentage change (1990–2021). UI, uncertainty interval; CI, confidence interval. SDI, Socio-demographic Index. Estimates based on food attribution fractions applied to GBD 2021 data.

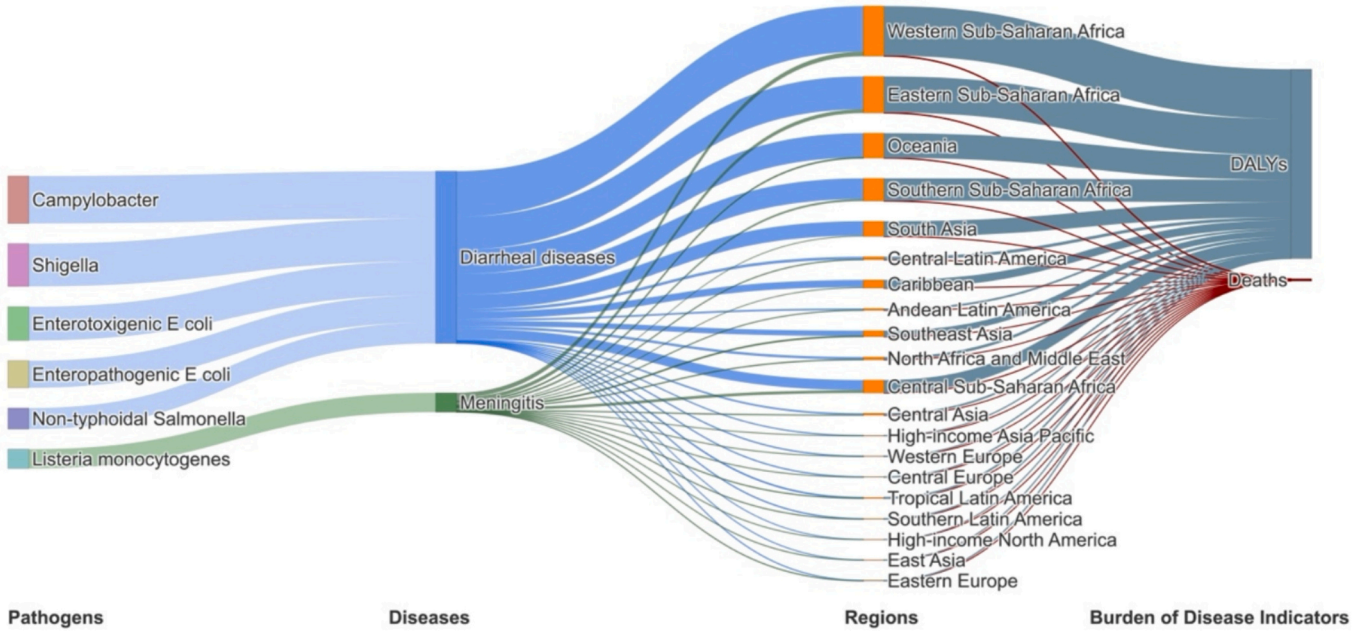


Fig. 1. Sankey diagram illustrating the flow of disease burden from pathogens to diseases, geographic regions, and DALYs associated with selected foodborne bacterial species (2021).

disproportionate enteric disease burden. In contrast, industrialized poultry production has emerged as the dominant exposure pathway for pathogens such as *Campylobacter* in high-income settings. Terminal flow convergence revealed that DALYs constituted the overwhelming majority of total burden, indicating that morbidity-driven disability rather than mortality alone defines the global foodborne disease landscape (Grace, 2015).

Spatiotemporal heat mapping (Fig. 2) revealed profoundly divergent regional trajectories across the 31-year analysis period. ASDALY (Fig. 2A) identified Western Sub-Saharan Africa as the most severely burdened region, with rates peaking at approximately 1469 per 100,000 in 1991. Although a substantial decline of approximately 74% was observed by 2021 (384 per 100,000), Western Sub-Saharan Africa retained the highest burden globally throughout the entire observation period, followed by Eastern Sub-Saharan Africa and South Asia. This trajectory underscores the limited capacity of incremental sanitation improvements to offset the cumulative burden imposed by persistent food system informality and constrained cold chain infrastructure in this region. High-income regions, by contrast, maintained consistently low rates across all study years. ASDR (Fig. 2B) displayed spatially concordant patterns. Western Sub-Saharan Africa similarly recorded peak mortality rates of approximately 17.36 per 100,000 in 1990, which declined to 4.73 per 100,000 by 2021, reflecting a comparable reduction of approximately 73%. The parallel magnitude of decline across both ASDALY and ASDR underscores the robustness of these temporal trends. Nevertheless, the persistence of marked regional disparities points to deeply entrenched structural inequities in food safety infrastructure, particularly the dominance of informal food markets comprising 70–80% of Sub-Saharan retail distribution, concurrent with limited cold chain coverage, constrained microbial testing capacity, and weak traceability systems (Grace, 2015; Grace et al., 2024; World Health Organization Regional Office for Africa, 2024).

Estimated annual percentage change mapping (Fig. 3) revealed epidemiological trajectories that challenge deterministic assumptions linking development status with disease control. Globally, the ASDALY for diarrheal diseases declined at an EAPC of −4.35% (95% CI: −4.67 to −4.03) from 1990 to 2021 (Table 2), with substantial regional heterogeneity in both magnitude and direction. Diarrheal diseases EAPC mapping (Fig. 3A-B) revealed a pronounced geographic dichotomy. Sub-Saharan Africa, South Asia, and Southeast Asia showed EAPC values ranging from −8% to −14%, indicating substantial burden reductions across multiple pathogens, with *Shigella*, *ETEC*, and *EPEC* all demonstrating declines. In contrast, several High-income nations exhibited divergent trajectories. Aggregated at the high-SDI quintile level, the ASDR EAPC for diarrheal diseases was +0.63% (95% CI: +0.13 to +1.12; Table 2), the only SDI stratum with a statistically significant positive mortality trend. While *Shigella*, *ETEC*, and *EP*EC consistently declined (EAPC: −2.5% to −5.8%), *Campylobacter* showed positive EAPC values, reflecting pathogen-specific emergence patterns.

Specifically, pathogen-specific analysis (Fig. S2) revealed marked divergence in transmission control across the development spectrum. In high-SDI nations, *NTS*, *ETEC*, and *EPEC* all declined substantially (*NTS*/*ETEC* EAPC: −4.2% to −6.5%; EAPC uniformly negative across all countries), reflecting the effectiveness of cold chain standardization and farm hygiene practices. By contrast, *Campylobacter* expanded from ~30% (1990) to ~60% (2021) of total diarrheal diseases burden in these regions, driven by incomplete post-harvest poultry control implementation. In low-SDI settings, *Shigella*, *ETEC*, and *EPEC* achieved marked burden reductions (EAPC: −6.6% to −12.8%), consistent with improved sanitation infrastructure and expanded surveillance. Notably, China exemplifies this transitional pattern: despite its middle-SDI status (SDI 0.69), the country achieved a diarrheal diseases EAPC of −12.80% through systematic food system modernization, including agricultural intensification, stringent regulatory oversight, and expanded cold chain penetration.

Aggregated at the high-SDI quintile level, the ASDR EAPC for diarrheal diseases was +0.63% (95% CI: +0.13 to +1.12; Table 2). This was the only SDI stratum showing a statistically significant positive mortality trend over the study period, driven predominantly by *Campylobacter* emergence. Interestingly, several high-income countries exhibited an increasing diarrheal diseases burden between 1990 and 2021 despite concurrent substantial meningitis declines, with the most pronounced increases observed in the Netherlands (+2.47%), Ireland (+2.08%), Denmark (+1.43%), France (+1.23%), Slovenia (+0.85%), and Switzerland (+0.64%; all $p < 0.001$; Data S1). The geographic clustering of these increases in Western and Northern Europe, alongside East Asia,

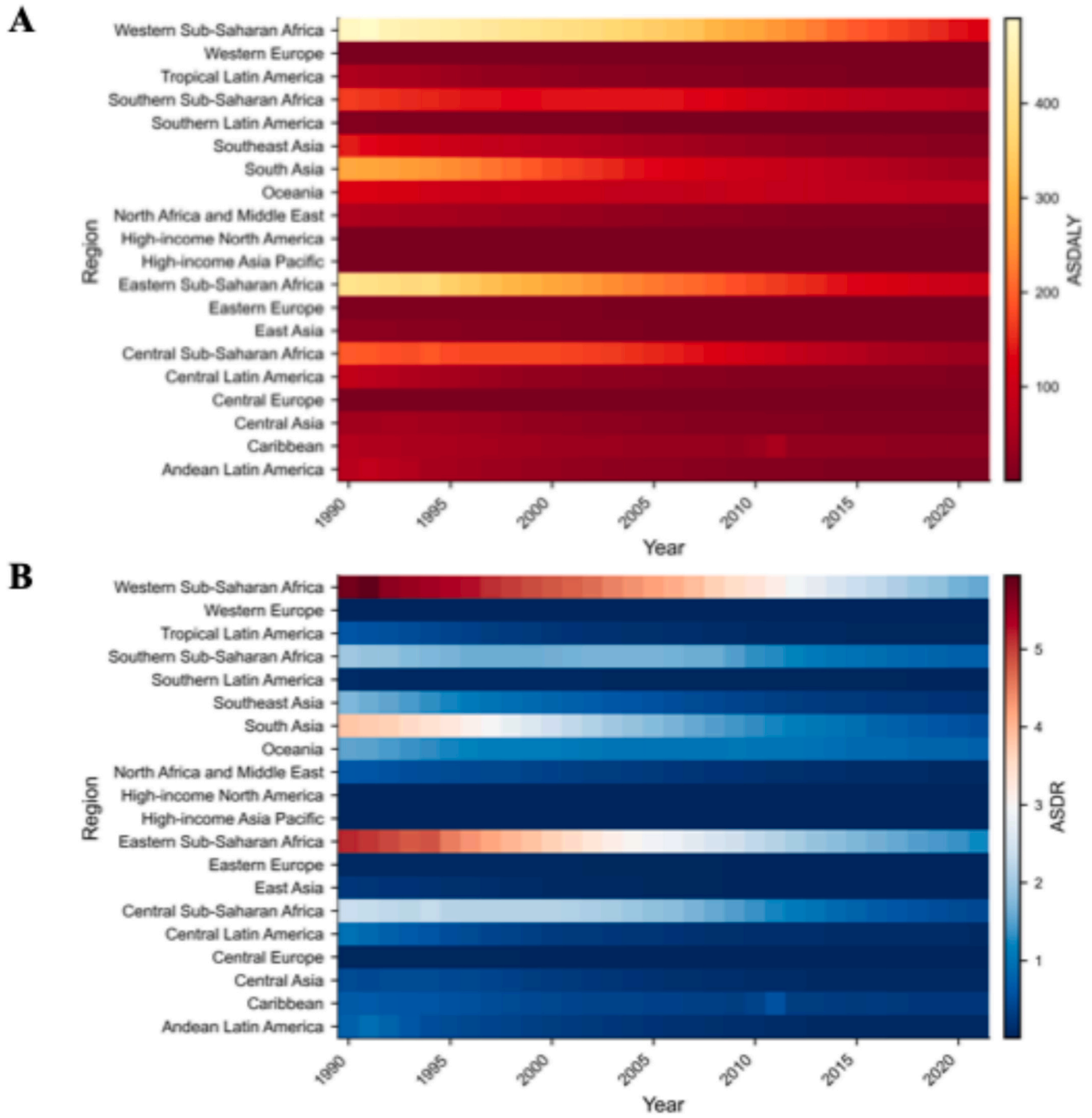


Fig. 2. Temporal and spatial trends in disease burden across 21 GBD geographic regions: (A) age-standardized disability-adjusted life year rate (ASDALY), and (B) age-standardized death rate (ASDR) from 1990 to 2021.

implicates shared structural features, particularly the co-expansion of intensive poultry industries with delayed regulatory responses, rather than country-specific idiosyncrasies. Pathogen-specific stratification revealed markedly different drivers: *Campylobacter* accounted for 60–70% (Data S1) of the observed increase in diarrheal diseases burden in these nations, while *NTS* and *ETEC* uniformly declined across all high-SDI countries (EAPC: −3.5% to −6.2%). The concordance between DALY-based and mortality-based EAPC estimates across these pathogens further validates the robustness of these directional trends (Fig. S3C). Similarly, *Shigella* and *EPEC* demonstrated consistently negative EAPCs across all SDI strata (EAPC: −2.5% to −5.8%, Fig. S3A–B). *EPEC* uniformly declined in all 15 exemplar countries examined, with no nation demonstrating emergence despite varying development status and food safety infrastructure maturity. This ubiquitous decline suggests a greater susceptibility to sanitation-based interventions, which reflects the dependence of enteropathogenic *EPEC* on direct fecal-oral transmission pathways, in contrast to the more complex animal-origin transmission ecology of *Campylobacter*. Notably, *Campylobacter* emergence in high-SDI nations reflects a temporal mismatch between antimicrobial stewardship policies (2005–2008) (Fig. S3D), and post-harvest processing infrastructure. Antimicrobial restrictions without concurrent biosecurity intensification inadvertently increased

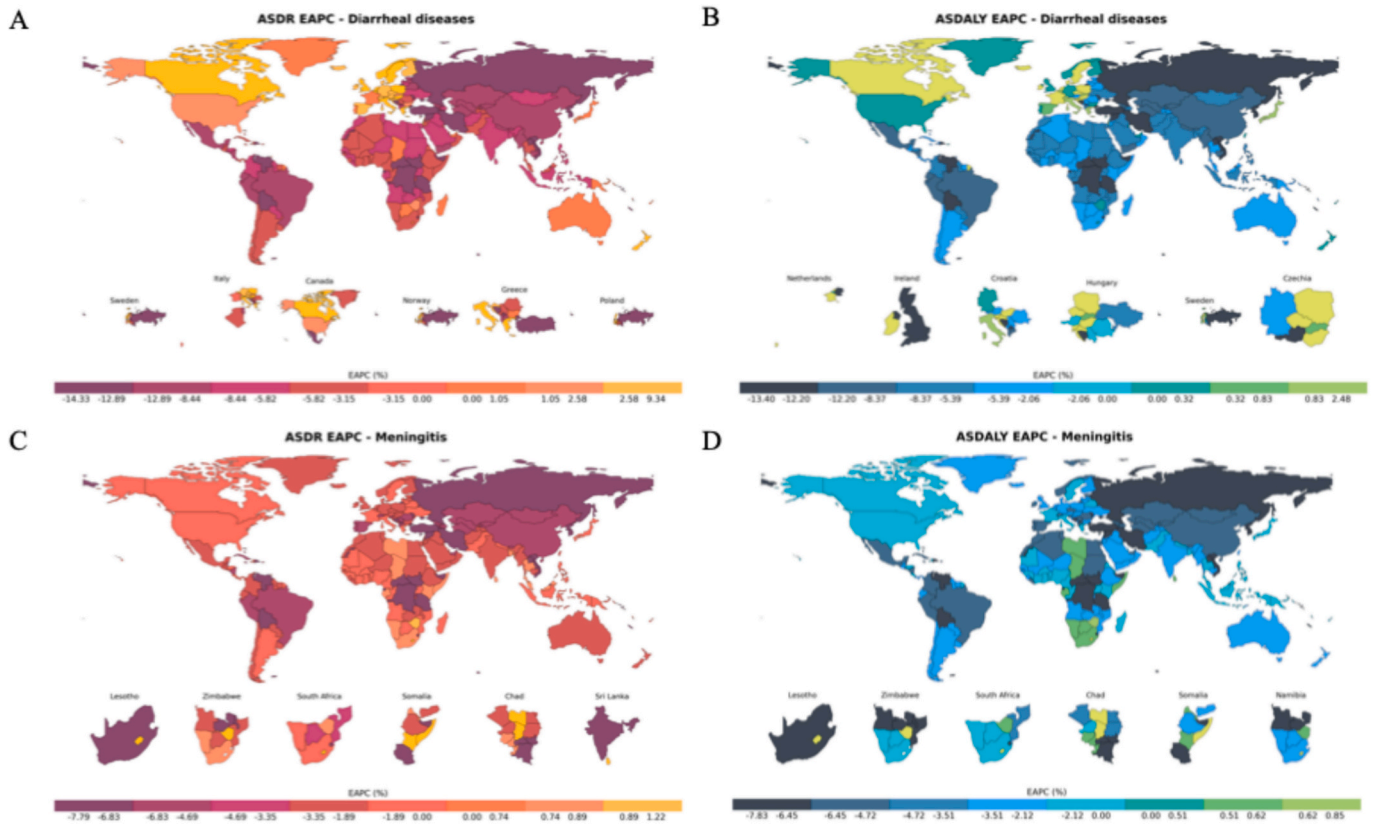


Fig. 3. Global geographic distribution of estimated annual percentage change (EAPC) for age-standardized disability-adjusted life years (ASDALY) and death rates (ASDR) across two disease categories: (A-B) diarrheal, (C—D) meningitis (1990–2021).

transmission risk in production systems with inadequate controls (Kaakoush et al., 2015; Newell et al., 2010). Conversely, *NTS* burden declined consistently across high-SDI countries despite poultry intensification, reflecting effectiveness of hazard analysis and preventive measures. Australia exemplifies integrated control: despite high poultry production, EAPC was −3.00% through systematic farm-to-consumer *Campylobacter* surveillance and mandatory end-product testing (Kirk et al., 2014).

In contrast, meningitis burden remained concentrated in Southern Sub-Saharan Africa (Fig. 3C-D), where country-level trajectories diverged sharply despite shared regional vulnerability. *L. monocytogenes* uniformly dominated meningitis across all 15 exemplar countries (Fig. S4), yet exhibited profound divergence in temporal trends stratified by SDI and population immunocompetence. High-income nations demonstrated consistent *L. monocytogenes* declines (Netherlands −4.05%, Ireland −4.83%, Denmark −4.65%, all $p < 0.001$), supported by robust cold chain systems and strong diagnostic capacity; conversely, Mauritius exhibited an anomalous increase (+4.41%, $p < 0.001$), creating an 8.8% EAPC divergence despite comparable economic metrics (Fig. S4H). This inversion reflects differential susceptibility mediated by population-level immunosuppression. The rising *L. monocytogenes* burden in Mauritius coincides with elevated HIV seroprevalence of approximately 2% of the population, which amplifies the risk of invasive infections associated with ready-to-eat food consumption (Barichello et al., 2023; Qu et al., 2023).

Within Southern Sub-Saharan Africa, Lesotho exhibited the most notable increase (+0.91%, $p < 0.01$) while South Africa demonstrated modest decline (−0.80%, $p < 0.01$), reflecting localized variations in HIV/AIDS-associated immunosuppression and diagnostic capacity. Across the examined development spectrum, these patterns demonstrated that food safety and health system infrastructure, rather than GDP per se, determine pathogen-specific meningitis burden trajectories, with the ~16-fold EAPC gradient across development levels (Figs. S2–S4) reflecting both structural capacity differences and population-level immunocompetence variations.

Collectively, these findings demonstrate that foodborne disease control trajectories are shaped primarily by food safety infrastructure capacity rather than development status alone. In high-income countries, incomplete implementation of farm-to-fork controls following antimicrobial stewardship policies (2005–2008) paradoxically generated *Campylobacter* emergence. In contrast, low-income regions show amplified meningitis burden driven by HIV-associated immunosuppression and demographic aging, despite improvements in sanitation and surveillance systems. Addressing these divergent trajectories requires context-specific interventions. In developed economies, this includes intensifying post-harvest processing and verifying cold-chain systems. For Sub-Saharan Africa, it involves expanding access to antiretroviral therapy and making targeted investments in food safety for informal market systems. These findings establish food safety infrastructure as fundamentally a development challenge inseparable from poverty reduction and nutritional security (Jaffee et al., 2018; Roesel and Grace, 2015).

3.2. Age and sex disparities in disease vulnerability

Age-sex pyramid analysis (Fig. 4) revealed a distinctive bimodal distribution for both diseases, with children under five and adults aged 70 years or older bearing a disproportionately high burden. For diarrheal diseases, age-specific DALY rates in children and elderly adults both exceeded 11,000 per 100,000—representing a 9-fold and 45-fold elevation relative to the global age-standardized rate of 312.95 per 100,000, respectively (Table 2). This pronounced pediatric vulnerability reflects the critical complementary feeding window at 6–23 months, when passive immunity from breastfeeding wanes and infants face

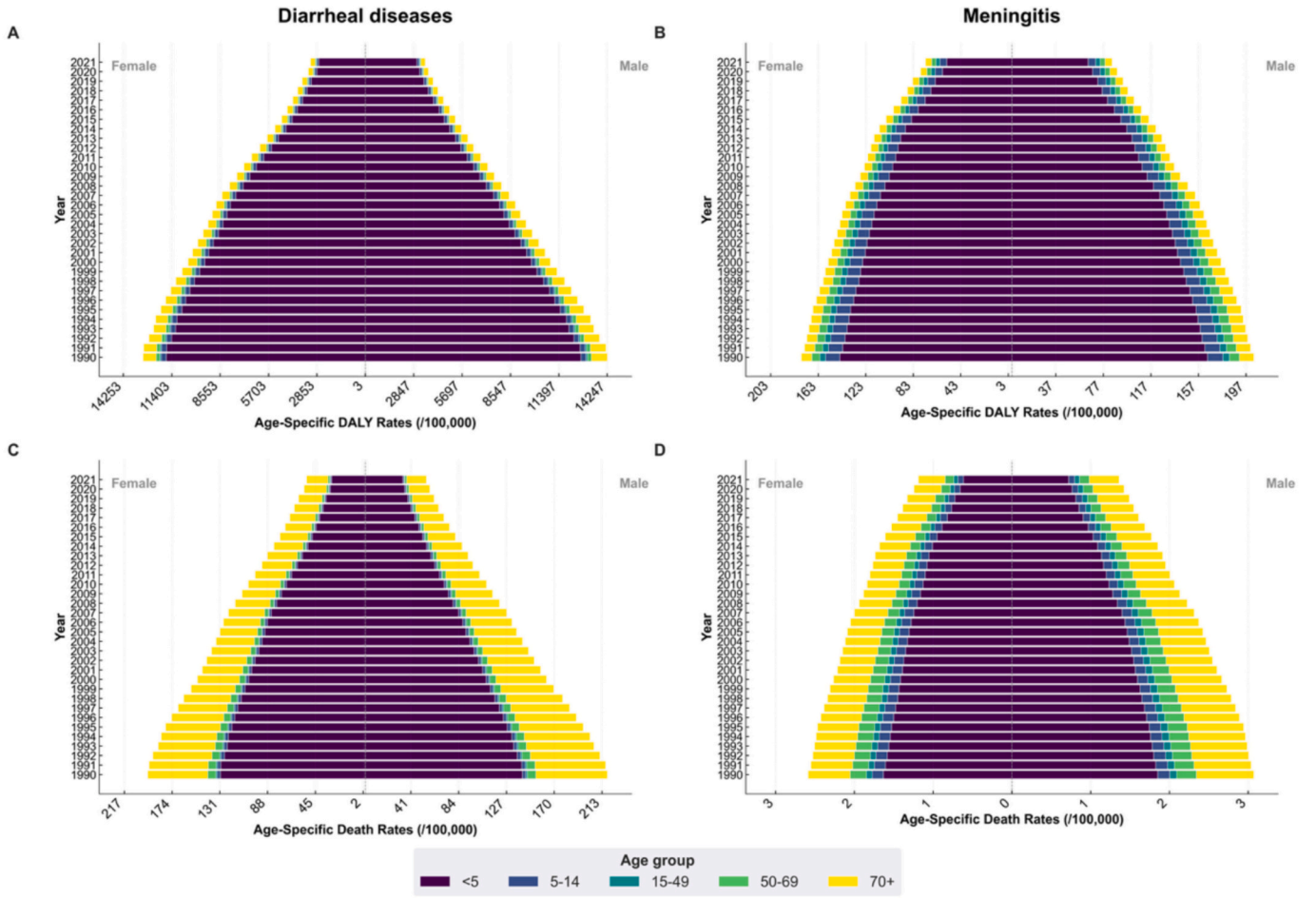


Fig. 4. Age-specific and sex-stratified disease burden for two disease categories: DALYs and mortality rates across five age groups (<5, 5–14, 15–49, 50–69, 70+ years) for diarrheal, and meningitis.

increased pathogen exposure via contaminated weaning foods (Kotloff et al., 2013). The elderly showed a distinct vulnerability profile, especially for invasive foodborne pathogens linked to meningitis. For *L. monocytogenes*, case-fatality rates rise from 1 to 3% in healthy adults to 20–30% in the elderly, driven by immunosenescence and dietary preferences for ready-to-eat refrigerated foods that enable psychrotrophic pathogen growth (Buchanan et al., 2017; Goulet et al., 2011). Bacterial meningitis followed a similar age-stratified trend, with a secondary burden peak in the elderly and an overall EAPC of −2.46 (95% CI: −2.69 to −2.23).

Sex-stratified analysis showed a consistent male predominance for both outcomes: males had 10.4% higher age-specific DALY rates for diarrheal diseases (327.95 vs. 296.98 per 100,000) and 16.1% higher rates for meningitis (9.54 vs. 8.22 per 100,000) (Table 2). These disparities were concentrated among working-age adults (15–49 years), likely reflecting occupational exposure in livestock and meat processing, plus behavioral factors such as higher raw meat consumption. This male predominance persisted across all SDI quintiles, though the magnitude of the sex gap varied by development level and did not follow a monotonic gradient, with a wider SDI-dependent range observed for meningitis than for diarrheal diseases (Havelaar et al., 2015). Notably, sex-specific vulnerability patterns remained stable over the study period despite large overall burden reductions, indicating sex-differential exposure mechanisms are not uniformly attenuated by socioeconomic progress. The enduring vulnerability of children and the elderly highlights the need for age-targeted food safety interventions: stricter microbiological standards for infant foods including formula and complementary products, and mandatory *L. monocytogenes* testing for ready-to-eat items commonly consumed by the elderly, who face sharply higher risks of invasive meningeal infection (Buchanan et al., 2017).

3.3. Socioeconomic gradients and food system determinants

The sociodemographic gradient analysis revealed an inverse relationship between development level and foodborne bacterial disease burden (Table 2). ASDALY steep stratification across SDI quintiles. For diarrheal diseases, Low-SDI regions had rates of 736.99 per 100,000 (195-fold excess compared to High-SDI settings at 3.78 per 100,000), while meningitis showed a 35-fold disparity (25.49 vs 0.73 per 100,000). The pathogen-specific analysis (Fig. 5) illustrated differential SDI vulnerabilities across all these pathogens. In the 2021 cross-section, Low-SDI countries sustained the highest aggregate foodborne DALY burden (1409.4 per 100,000), predominantly driven by *Shigella* (514.1 per 100,000), *ETEC* (280.9 per 100,000), and *EPEC* (181.9 per 100,000), reflecting fecal-oral transmission due to inadequate cold chain infrastructure and unhygienic food handling in low-resource settings. Low-middle-SDI countries exhibited intermediate aggregate burden (881.3 per 100,000), with *Shigella* (319.4 per 100,000) and *ETEC* (193.4 per 100,000) remaining the dominant contributors, whereas Middle-SDI countries showed substantially attenuated burden (207.4 per 100,000). Conversely, High-SDI nations had minimal burden, with residual risk in animal-source food supply chains and ready-to-eat products, consistent with documented hazards in poultry and eggs.

Spearman correlation analyses examined food system indicators from two dimensions. Associations between food system indicators and total DALY are presented in Fig. 6, counts across 1941 country-year

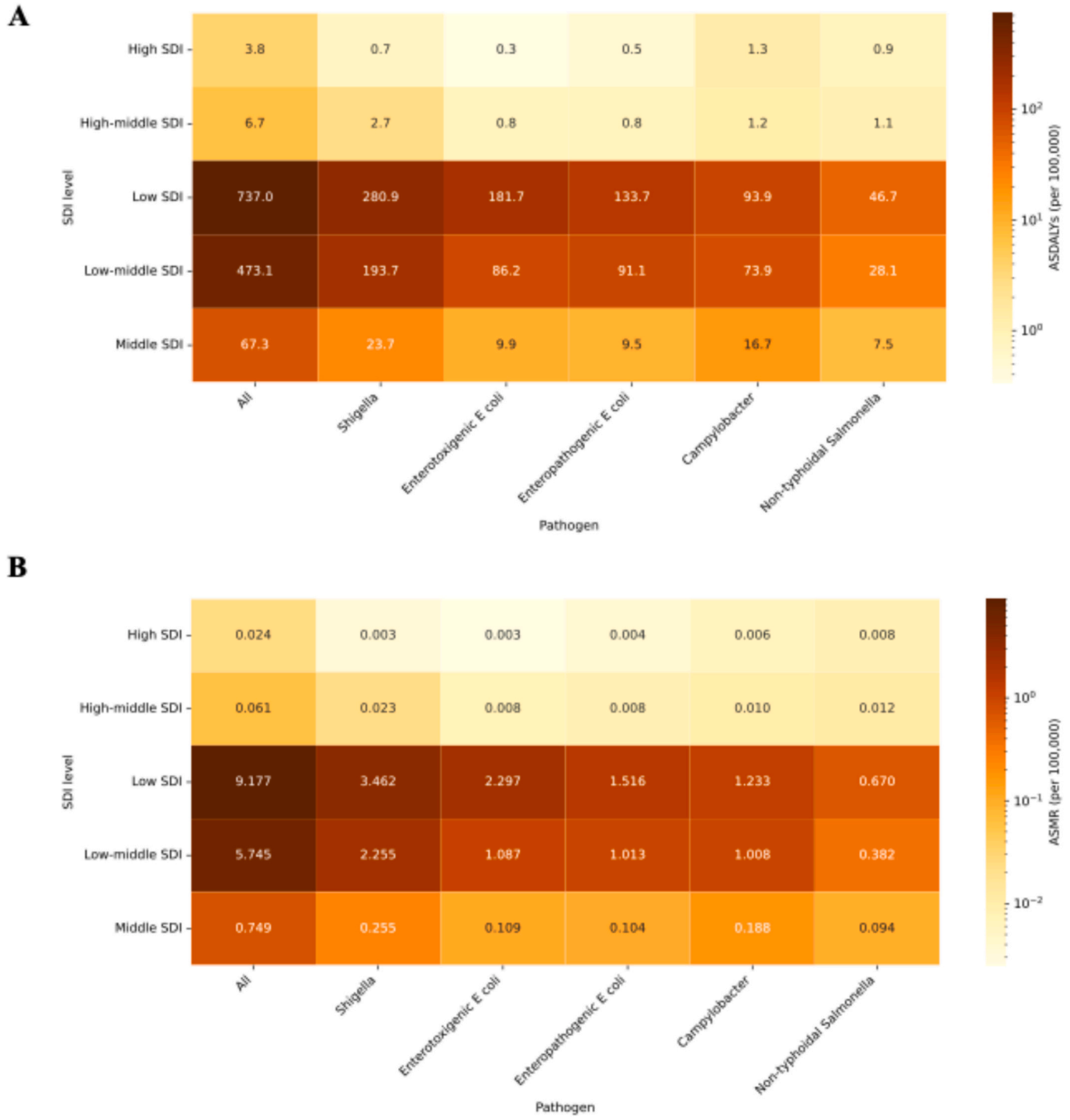


Fig. 5. ASDALY and ASDR of foodborne diseases by SDI level and pathogen.

observations (1990–2021), capturing aggregate population-level burden pressure. Table 3 reports correlations with age-standardized DALY rates (ASDALY per 100,000), enabling per-capita risk comparisons independent of population size and age structure. Regarding absolute DALY burden (Fig. 6A–B), vegetable production showed the strongest positive associations across syndromes (diarrheal diseases: $\rho = +0.48$; meningitis: $\rho = +0.55$), followed by fruit production (diarrheal diseases: $\rho = +0.27$), while meat production exhibited weak and largely non-significant positive associations (diarrheal diseases: $\rho = +0.02$). SDI and GDP per capita had weak associations with total DALY counts. These patterns reflect a population-scale co-distribution effect: countries with the largest absolute DALY counts are predominantly large-population, low-SDI settings in South Asia and Sub-Saharan Africa, where subsistence-oriented vegetable and legume farming co-exists with high absolute disease counts driven by inadequate post-harvest hygiene, contaminated irrigation water, and limited cold chain infrastructure (A. Corina Fărcaș, 2024; Los et al., 2018). Within the matrices, strong positive inter-predictor correlations among production variables are also apparent (meat vs. cereal production: $\rho = 0.82$; fruit vs. cereal: $\rho = 0.73$; vegetable vs. meat: $\rho = 0.56$), reflecting the co-development of diversified agricultural sectors in middle- and high-income economies, and should not be interpreted as production-to-burden associations.

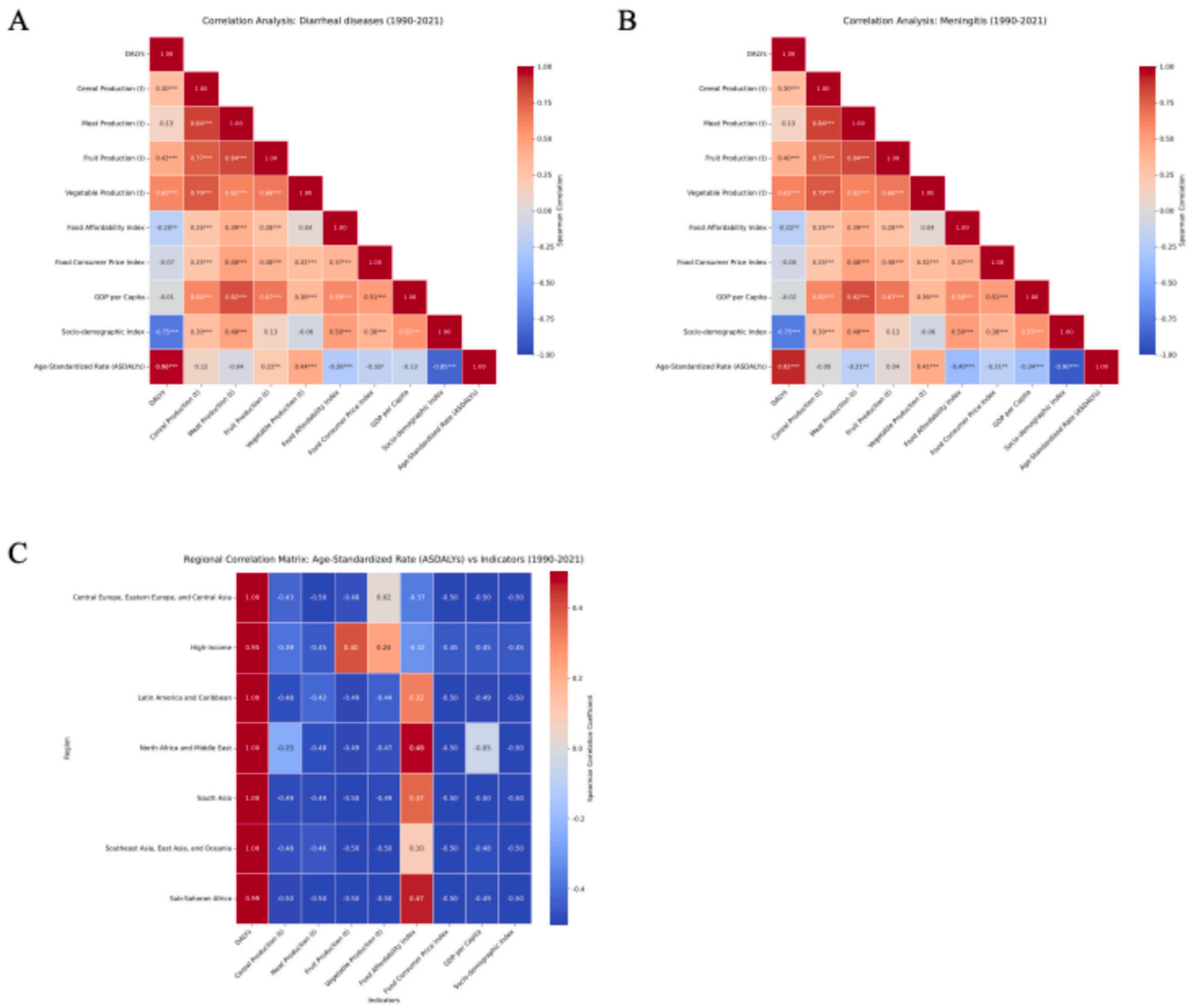


Fig. 6. Correlation heatmaps of feature relationships: spearman correlation coefficients between socioeconomic, food system indicators, and age-standardized disease burden metrics. (A) diarrheal diseases. (B) meningitis. (C) regional.

Regional analysis (Fig. 6C) corroborated these patterns: Sub-Saharan Africa exhibited positive correlations between absolute DALY burden and production indicators ($\rho \approx +0.17$–0.29), while high-income regions showed near-zero or weakly negative associations ($\rho \approx -0.11$ to -0.13), consistent with the SDI-stratified gradient documented in Table 2.

Per-capita analysis via age-standardized rates (Table 3) reveals the structural determinants of individual-level foodborne disease risk. SDI emerged as the dominant protective factor across both disease categories ($\rho = -0.830$ overall; diarrheal diseases: $\rho = -0.832$; meningitis: $\rho = -0.905$). As a composite developmental metric integrating income, education, and fertility, SDI reflects a country's underlying capacity to sustain food safety infrastructure investment. The mechanistic basis of this association becomes evident when SDI is examined alongside production-level data. SDI demonstrated strong positive correlations with animal-source food production (egg: $\rho = +0.593$; meat: $\rho = +0.601$; dairy: $\rho = +0.559$; Table S3), and yet these same production categories were inversely associated with age-standardized burden (egg: $\rho = -0.277$; meat: $\rho = -0.272$; dairy: $\rho = -0.224$; Table 3). This structural inversion reflects development-mediated food safety capacity, whereby high-SDI countries deploy comprehensive HACCP protocols, cold chain infrastructure, and pathogen reduction technologies that decouple production scale from per-capita transmission risk (Jaffee et al., 2018). In LMICs where production intensification outpaces regulatory capacity, however, this protective inversion does not materialize. The positive association between legume production and burden in low-SDI settings (Table 3) illustrates this point, suggesting that expansion without commensurate infrastructure investment may instead amplify contamination risk (Nyokabi et al., 2023). GDP per capita and food affordability reinforced this developmental gradient, each exhibiting moderate negative correlations with age-standardized burden ($\rho = -0.362$ and $\rho = -0.265$, respectively; Table 3). These associations are consistent with income growth facilitating investment in cold chain logistics, formalized retail systems, and enhanced microbiological controls. Plant production indicators presented a contrasting risk profile. Legume production showed consistent positive associations with ASDALYs across both syndromes (overall: $\rho = +0.406$; diarrheal diseases: $\rho = +0.424$; meningitis: $\rho = +0.397$; Table 3). Vegetable production similarly demonstrated a positive association, though of more modest magnitude ($\rho = +0.198$; Table 3). This pattern likely reflects the concentration of legume-based subsistence farming in low-SDI settings,

Table 3
Network pathway correlations for foodborne disease burden determinants.

Variable category	Variable	Overall ASDALY (ρ)	Diarrheal (ρ)	Meningitis (ρ)
Socioeconomic indicators				
	Socio-demographic Index	−0.830***	−0.832	−0.905
	Urbanization rate	−0.371***	−0.349	−0.508
	GDP per capita	−0.362***	−0.339	−0.504
Food price indicators				
	Food affordability	−0.265***	−0.257	−0.362
	Food consumer prices	−0.190***	−0.181	−0.260
	General consumer prices	−0.178***	−0.170	−0.249
	Food price inflation	NS	−0.053	−0.115
	Food to General Price Ratio	NS	−0.112	−0.171
	Food price volatility	NS	−0.082	−0.139
Animal product production				
	Egg production	−0.277***	−0.261	−0.382
	Meat production	−0.272***	−0.256	−0.367
	Dairy production	−0.224***	−0.200	−0.321
Plant product production				
	Legumes production	+0.406***	+0.424	+0.397
	Vegetables production	+0.198***	+0.210	+0.206
	Fruits production	NS	+0.020	−0.102
	Cereals & grains production	NS	−0.073	−0.148
Trade patterns (protective)				
	Cereals & grains net export	−0.339***	−0.346	−0.283
	Dairy net export	−0.288***	−0.300	−0.194
	Eggs net export	−0.266***	−0.268	−0.226
	Legumes net export	−0.194***	−0.194	−0.158
Trade patterns (risk)				
	Nuts net export	+0.413***	+0.409	+0.422
	Vegetables net export	+0.238***	+0.237	+0.167
	Fruits net export	+0.228***	+0.213	+0.249
	Meat net export	NS	+0.003	+0.019
Disease syndromes				
	Diarrheal	+0.999***	–	–
	Meningitis	+0.970***	–	–

Notes: Syndrome-specific correlations. ρ = Spearman correlation coefficient. ***p < 0.001, **p < 0.01, *p < 0.05 (FDR-adjusted). NS = not significant in overall ASR model but included for syndrome-specific associations. Negative correlations indicate protective associations; positive correlations indicate risk associations.

where limited post-harvest hygiene infrastructure amplifies fecal-oral transmission.

Trade patterns further differentiated protective from risk-associated pathways. Net exports of cereals (ρ = −0.339), dairy (ρ = −0.288), and eggs (ρ = −0.266) were each inversely associated with burden, consistent with export-oriented systems operating under more stringent international sanitary standards. By contrast, net exports of nuts (ρ = +0.413), vegetables (ρ = +0.238), and fruits (ρ = +0.228) showed positive risk associations, suggesting that high-volume perishable supply chains may amplify contamination exposure where post-harvest cold chain infrastructure remains underdeveloped. These findings suggest that food safety infrastructure investment, rather than production volume, is a more modifiable determinant of per-capita foodborne disease risk.

Temporal analysis of normalized disease burden and food system indicators across eight global regions (Fig. 7A–H) provided longitudinal evidence broadly consistent with the cross-sectional Spearman findings. In high-income regions (Fig. 7B), normalized ASDALY declined steeply toward near-zero while production indices for meat, cereals, eggs, and dairy continued to rise throughout 1990–2021. This trajectory exemplifies production-safety decoupling, whereby improvements in cold chain coverage, formalized retail channels, and microbiological controls reduce transmission irrespective of production scale (Kowalcyk et al., 2025). Central Europe, Eastern Europe, and Central Asia (Fig. 7A) exhibited a comparable pattern, with burden declining sharply from 1990 onward alongside sustained production growth. North Africa and the Middle East (Fig. 7D) similarly demonstrated progressive burden reduction concurrent with rising production activity. South Asia (Fig. 7E) and Southeast Asia, East Asia, and Oceania (Fig. 7F) displayed intermediate trajectories, with substantial burden decline occurring alongside rapid production intensification. These patterns are consistent with early-stage food safety transition driven by urbanization (a significant protective factor, ρ = −0.371, Table 3) and formal market expansion. Latin America and the Caribbean (Fig. 7C) occupied a broadly comparable intermediate position, suggesting ongoing but incomplete infrastructural transition. Sub-Saharan Africa (Fig. 7G) diverged markedly from these regional trends. Despite steady production increases throughout the observation period, normalized burden declined at a substantially slower rate, reinforcing that production expansion unaccompanied by infrastructure investment is insufficient to achieve comparable burden reductions (Sheehama and Singh, 2025). Across all eight regions, the global summary panel (Fig. 7H) shows a broad co-decline of normalized ASDALY alongside rising SDI and GDP per capita. Combined with the cross-sectional correlations in Table 3, this longitudinal pattern suggests that food safety infrastructure development, mediated through broader socioeconomic advancement, may be a more important driver of population-level burden reduction than agricultural output volume alone.

3.4. Network analysis: multi-pathway burden determinants

Pathway network analysis integrated socioeconomic indicators, food production and trade systems, pathogen transmission routes, and disease-specific outcomes across 1941 country-years (Fig. 8; Table 3). Variable importance ranking (Fig. S5; Table S3) confirmed the hierarchical structure identified, with SDI functioning as the central hub node connecting food production systems to health outcomes. Beyond the cross-sectional SDI–burden gradient established above, network topology revealed syndrome-specific architecture. SDI node connections to *L. monocytogenes* burden were stronger than those to enteric syndromes, suggesting that cold chain dependency and ready-to-eat food pathways are more tightly governed by development-level infrastructure than fecal-oral routes (Jaffee et al., 2018; Unnevehr and Ronchi, 2015). Legumes exhibited an inverse configuration, with SDI negatively correlated with production (ρ = −0.153; Table S3) while legume production positively correlated with disease burden (ρ = +0.406; Table 3), positioning legume-associated pathways as a distinct low-SDI vulnerability node. However, differences in legume varieties, preparation methods, or surveillance sensitivity may account for this pattern.

The divergent patterns between legume production and legume trade may reflect quality certification requirements for export markets. Exported legumes undergo mandatory microbiological testing and moisture control, reducing contamination risks. Conversely, domestically marketed legumes in low-SDI countries often bypass such controls, with prolonged storage at ambient temperature facilitating bacterial proliferation (particularly Shigella and EPEC in legume-based dishes).

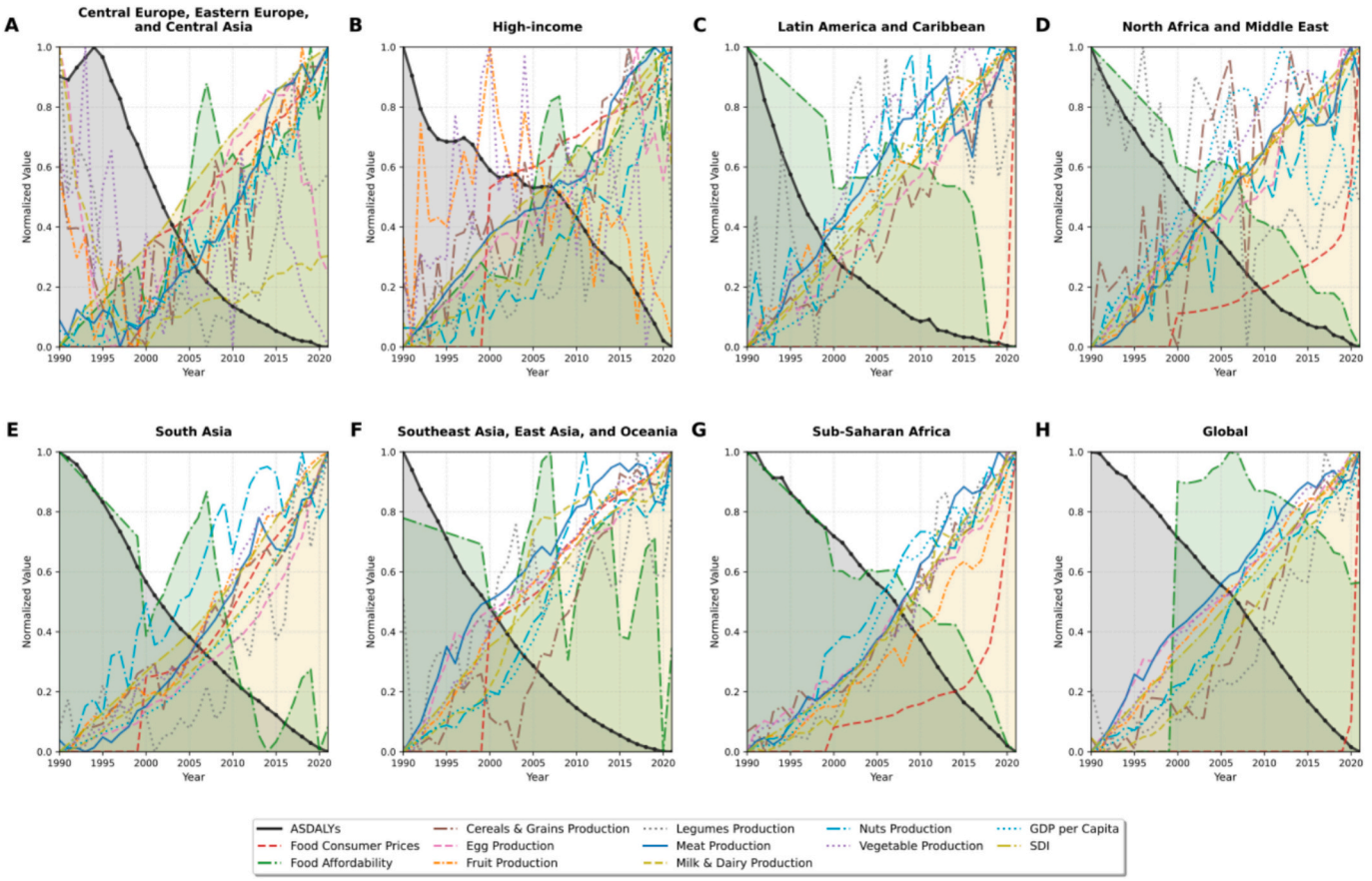


Fig. 7. Temporal trends of disease burden and food system indicators across regions (1990–2021).

Pathogen-level correlations between food indicators and pathogen-specific foodborne burden further indicated development-mediated decoupling between production scale and per-capita risk. Pathogen-specific correlations with food production variables (derived using pathogen-specific foodborne burden estimates per Section 2.1.2) are presented in Supplementary Fig. S6, revealing consistent inverse associations with animal-sourced food production: *Campylobacter* with meat ($\rho = -0.383$) and egg products ($\rho = -0.377$); *NTS* with egg ($\rho = -0.309$) and meat production ($\rho = -0.291$); and *L. monocytogenes* with egg ($\rho = -0.382$), meat ($\rho = -0.367$), and dairy production ($\rho = -0.321$). These inverse associations likely reflect food safety capacity in high-output systems, where veterinary surveillance, processing controls, and cold-chain coverage can interrupt transmission despite increased production volumes (Havelaar et al., 2015). Conversely, plant-associated enteric pathogens showed positive correlations with corresponding food categories. Among the pathogens examined, *Shigella*, *EPEC*, and *ETEC* all showed positive correlations with vegetable and legume production (*Shigella*: $\rho = +0.326, +0.544$; *EPEC*: $\rho = +0.174, +0.389$; *ETEC*: $\rho = +0.296, +0.455$). Notably, positive correlations with legume production were consistent across all six pathogens, including *NTS* ($\rho = +0.388$), *Campylobacter* ($\rho = +0.350$), and *L. monocytogenes* ($\rho = +0.397$), positioning legume farming systems as a cross-pathogen vulnerability node rather than a fecal–oral-specific pattern. These divergent patterns suggest that fecal–oral transmission amplification in low-SDI agricultural systems may be linked to irrigation with untreated water, uncomposted manure application, and limited post-harvest sanitization, with different relationships between production intensification and food safety outcomes across commodity types (Machado-Moreira et al., 2019).

International trade patterns exhibited systematic asymmetries that encode global disparities in food safety capacity. Nut, fruit, and vegetable net exports showed positive correlations with disease burden (overall ASR: $\rho = +0.347, +0.193$, and $+0.177$, respectively; Table S4), commodities predominantly from low- and middle-SDI regions. This pattern potentially reflects dual-standard quality systems, in which export-grade products undergo more stringent quality control while domestically marketed products remain subject to limited regulatory oversight, thereby concentrating contamination risks within producer-country populations (Gow and Cocks, 2007). Conversely, cereals, dairy, and egg net exports showed negative associations with disease burden ($\rho = -0.280, -0.226$, and -0.219, respectively; Table S4). These commodities are predominantly exported by high-SDI nations where private food safety standards and mandatory end-product testing reduce pathogen transmission risk across both domestic and export supply chains (Henson and Humphrey, 2010). Furthermore, cereals and dairy products undergo processing steps such as milling and heat treatment that limit microbial survival, while egg-exporting nations generally maintain biosecurity protocols including regulated flock vaccination and systematic *Salmonella* control programmes (FAO, 2023). These observations suggest that both export capacity and disease burden reflect broader developmental factors, with alternative explanations including differences in crop varieties, handling practices, or surveillance capacity between exporting and importing countries. The asymmetric association between commodity type and disease burden thus reflects the dual-standard dynamic prevalent in low- and middle-SDI settings, wherein export-oriented supply chains are selectively upgraded to meet importing-country sanitary requirements while domestic informal market chains remain insufficiently regulated (Grace et al., 2024; Kuboka et al., 2024).

Plant-associated enteric pathogens (*Shigella*, *EPEC*) demonstrated contrasting trade–production patterns. While vegetable and legume production correlated positively with disease burden in low-SDI

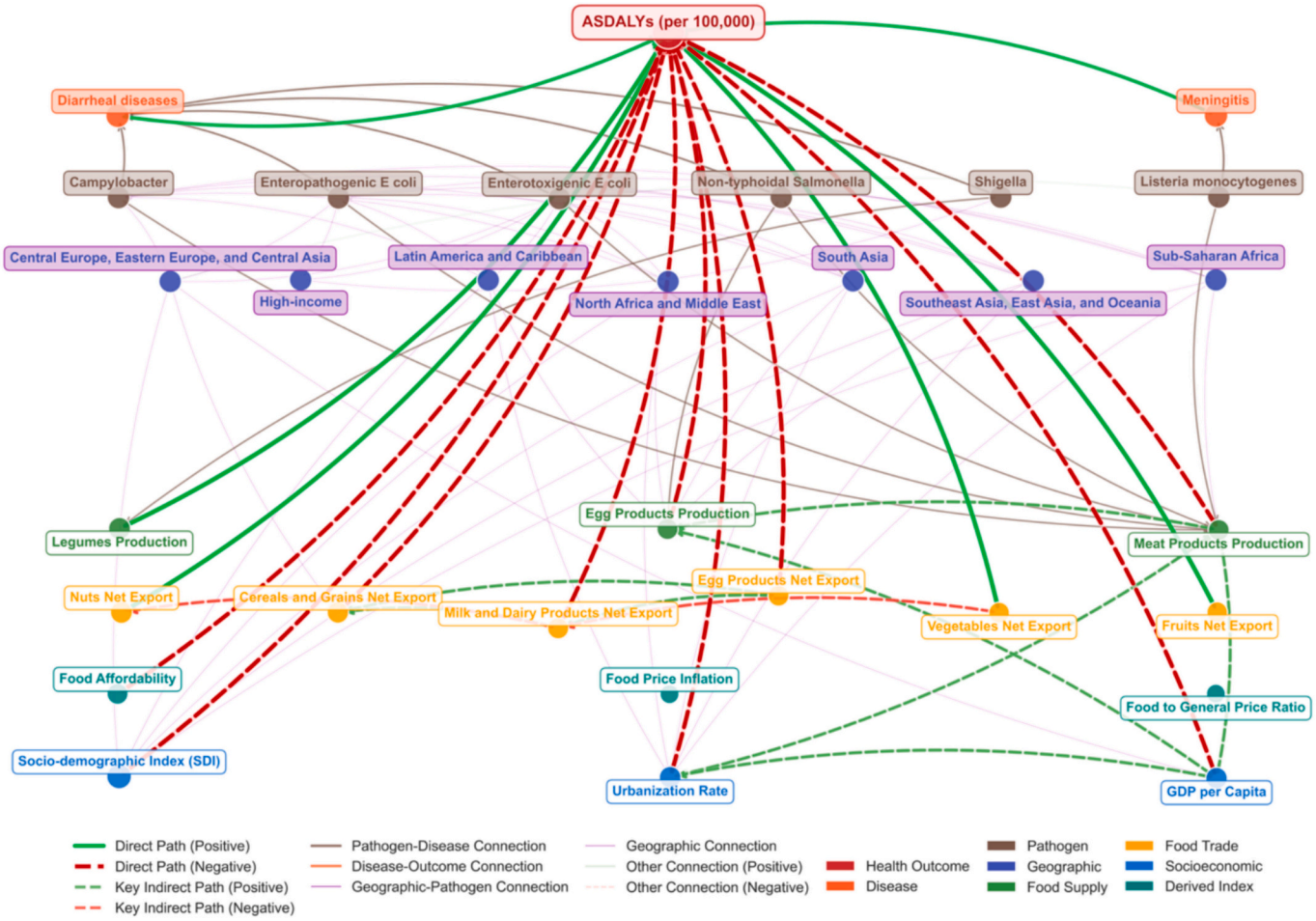


Fig. 8. Critical pathway network of foodborne disease burden and associated determinants.

contexts, vegetable net exports were negatively associated with overall burden ($\rho = -0.177$; Table S4), consistent with stricter pre-export microbial testing and water safety standards. This pattern suggests that commodity destination (export vs. domestic) may serve as a proxy for regulatory intensity, particularly for fecal–oral transmission routes vulnerable to irrigation and handling practices.

Syndrome-specific analysis revealed differential correlation strengths across disease categories (Table S4). meningitis showed stronger negative correlations with GDP per capita ($\rho = -0.504$) compared to diarrheal diseases ($\rho = -0.339$), suggesting that invasive foodborne infections may be more sensitive to health-system capacity than enteric syndromes. Basic water and sanitation investments, while reducing diarrheal diseases burden, may prove insufficient for invasive infections without higher-order health system strengthening. Alternative explanations, including differential case ascertainment and surveillance completeness, cannot be excluded from aggregate-level data.

3.5. Long-term burden projections under current trajectory scenarios

Time-series extrapolation based on regional data from 1990 to 2021 projected foodborne bacterial burden through 2050 under a continuation-of-trend assumption, and indicated substantial heterogeneity across regions (Fig. 9). At the global level (Fig. 9A), age-standardized burden is projected to decline for both syndromes. In 2021, the global ASDALY was 312.95 per 100,000 for diarrheal diseases (95% UI 164.59 to 542.73) and 8.90 per 100,000 for meningitis (95% UI 6.41 to 11.88). Extrapolation to 2050 suggests continued decreases in the central estimates. By 2050, central projections indicate a 69.50% decline in diarrheal disease ASDALY (from 312.95 to ~95.45 per 100,000) and a 64.59% decline in meningitis ASDALY (from 8.90 to ~3.15 per 100,000) relative to 2021 estimates (Fig. 9A). Uncertainty intervals widen progressively with forecast horizon, reflecting the accumulation of prediction error when models are extended beyond the temporal range of the training data. This behaviour is a general statistical property of long-term time-series forecasting rather than a consequence of specific causal assumptions (Balta et al., 2024). These projections should therefore be interpreted as plausible trajectories conditional on historical rates of improvement, rather than as deterministic outcomes.

Regional patterns (Fig. 9 B1-B2 and C1-C7) highlight pronounced divergence in projected trajectories. High-income regions (Fig. 9 C2) show the steepest declines in absolute ASDALY rates, specifically the largest absolute reductions in projected ASDALY per 100,000. Diarrheal diseases and meningitis approaching low-level plateaus by 2050, consistent with sustained risk reduction under mature food safety regulation and strong public health capacity. In contrast, Sub-Saharan Africa (Fig. 9 C4) remains the highest-burden region across the projection period and exhibits wider uncertainty bands, indicating that existing regional disparities are unlikely to narrow substantially if historical improvement rates persist. Even by 2050, Sub-Saharan Africa's projected diarrheal disease burden remains 51.66-fold higher than high-income regions (79.84 vs. 1.55 per 100,000), with meningitis burden 29.73-fold higher (9.34 vs. 0.31 per 100,000), highlighting persistent structural inequities even under optimistic assumptions (Fig. 9C2, C4). This pattern aligns with structural constraints including the predominance of informal food markets, limited cold-chain penetration, and

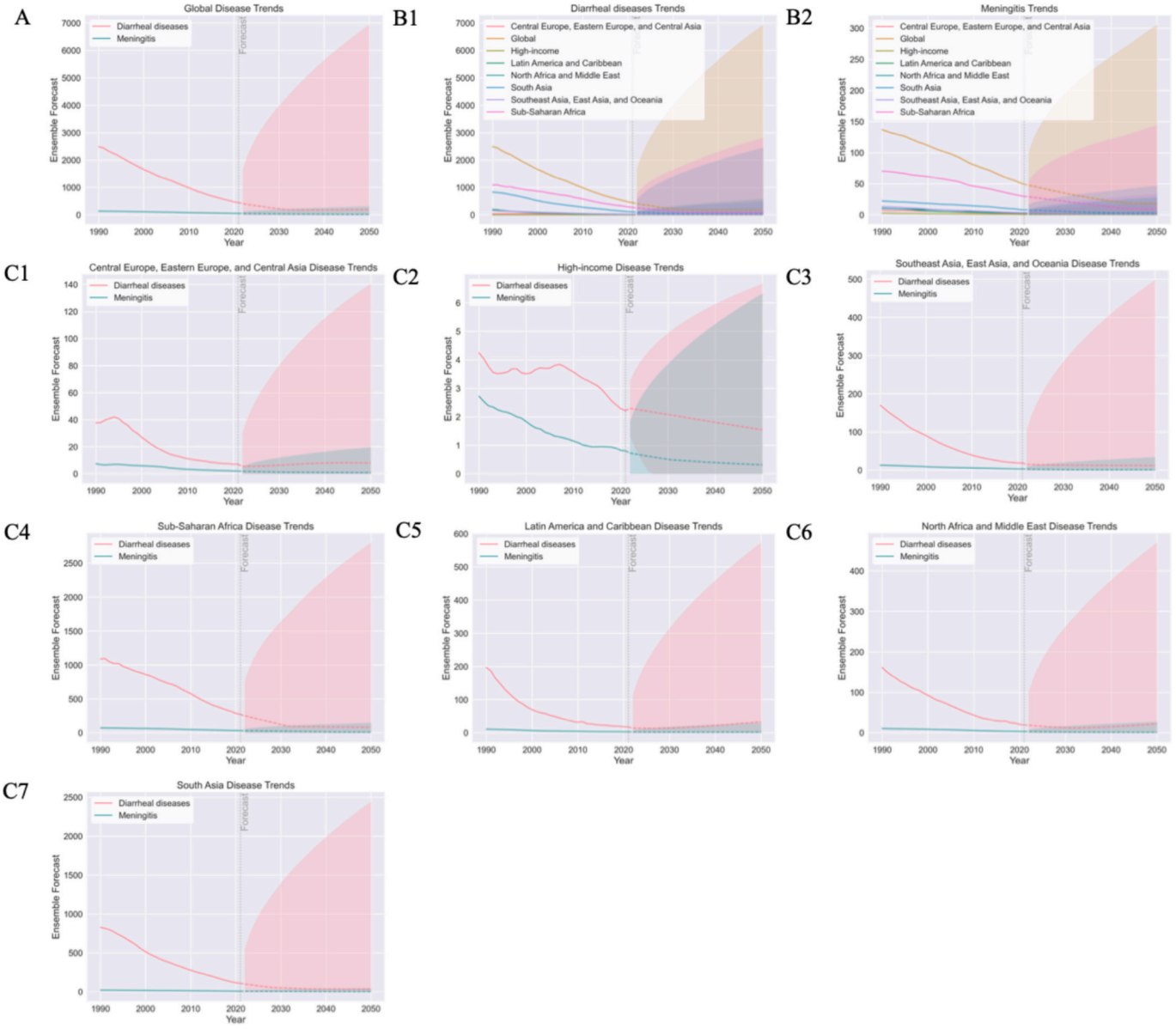


Fig. 9. Predictive modeling of future disease burden trends: forecasted age-standardized disability-adjusted life year rates (ASDALY) from 2022 to 2050 based on historical data (1990–2021).

insufficient veterinary service coverage and surveillance capacity (Sheehama and Singh, 2025). South Asia (Fig. 9 C7) displays an intermediate trajectory, with continued declines but persistently elevated absolute burden, which may reflect rapid intensification of livestock and food production outpacing the development of regulatory and enforcement capacity (Seyoum et al., 2024).

These continuation-of-trend scenarios do not account for potential non-linear disruptions, including climate-driven shifts in pathogen ecology, accelerated dissemination of antimicrobial resistance, extreme weather events, or transformative innovations in food systems and governance (Kounatidis et al., 2025). The marked expansion of uncertainty intervals after 2035, particularly in high-burden regions, should not be interpreted as evidence of an inevitable reversal in progress. Instead, it indicates that long-term extrapolation becomes increasingly uncertain in the absence of information on future interventions and structural change. A sensitivity analysis excluding high-SDI countries confirmed that projected absolute burden levels are substantially higher when high-performing nations are removed (diarrheal diseases: 3.80 vs. 3.30 per 100,000 by 2050; meningitis: 0.29 vs. 0.27), while the directional trend of continued decline remained consistent, supporting the robustness of the qualitative conclusions (Fig. S7). Overall, the projections are most useful for quantifying the urgency of action and identifying priority regions, and they reinforce the need for accelerated and targeted food safety capacity building in low-resource settings to alter the expected trajectory under historical trends.

3.6. Limitations

Several methodological constraints should be considered when interpreting these findings. As an ecological analysis of national-level data across 204 countries, all associations are observational and subject to the ecological fallacy; population-level correlations cannot be taken as evidence of individual-level causal pathways. The FAFs were derived from WHO expert elicitation (2010–2015) and applied uniformly across 1990–2021, potentially failing to capture temporal shifts in transmission driven by cold-chain expansion or retail modernization, particularly for multi-route pathogens such as *Shigella* and *EPEC*; applying super-region-level FAFs to constituent countries may further

obscure within-country heterogeneity in food safety infrastructure. GBD 2021 burden estimates also vary in precision across settings, with underreporting in low-resource environments likely producing systematic underestimation. Time-series projections assume historical trend continuity, and non-linear disruptions from climate change, antimicrobial resistance, or major policy shifts remain unmodeled; long-term forecasts should therefore be interpreted as conditional trajectories.

A further limitation concerns the exclusion of *Staphylococcus aureus* (*S. aureus*). Although WHO-FERG assigns a high food attribution fraction (FAF = 0.98) to staphylococcal intoxication, GBD 2021 allocates *S. aureus* burden to lower respiratory tract infections (LRTI) and meningitis rather than to the toxin-mediated gastroenteritis that constitutes the primary foodborne concern. This pathway mismatch precludes extraction of a directly comparable attributable burden estimate without substantial misclassification risk, necessitating exclusion from the present analysis. This classification gap underscores the need for harmonized disease frameworks in future GBD iterations.

Despite these constraints, consistent association patterns across all six pathogens and both disease syndromes, combined with good concordance between cross-sectional and longitudinal analyses, support the robustness of findings. The pathogen–food vehicle associations identified are biologically plausible, and few prior studies have integrated pathogen-specific, food-attributable burden estimates with food system dynamics at comparable geographic scale. Future research should prioritize temporally dynamic FAFs, subnational burden estimation, and longitudinal cohort validation of the ecological associations reported here.

4. Conclusion

By combining pathogen- and disease-specific food attribution fractions with GBD 2021 data and FAO food system indicators, this study quantified the global foodborne burden of six priority bacterial pathogens from 1990 to 2021. These pathogens resulted in an estimated 364,500 deaths from diarrhea (ASDR 3.63/100,000) and 11,100 from meningitis (ASDR 0.12/100,000) in 2021, with the burden mainly concentrated in Sub-Saharan Africa and South Asia. This pattern is consistent with a significant mismatch between the increase in agricultural production and the development of food safety infrastructure. Importantly, pathogen–food vehicle analysis suggests that the associations between production and disease burden depend on the maturity of the food safety system: animal-source food production appears protective in high-SDI countries with established HACCP systems, cold chain coverage, and pathogen reduction technologies. Conversely, in low-SDI settings, pathogens associated with plants (*Shigella*, *EPEC*) show increased risk, potentially associated with inadequate post-harvest sanitation, untreated irrigation water, and limited microbiological monitoring. This systematic reversal of pathogen–vehicle transmission patterns across different levels of development is consistent with a food safety transition framework, suggesting that production scale alone may not necessarily be the primary determinant of population health risks.

For the food industry, these findings highlight several key intervention priorities. Improved *L. monocytogenes* testing for ready-to-eat products aimed at elderly populations, stricter microbiological standards for infant complementary foods, and integrated farm-to-fork control of *Campylobacter* and *Salmonella* in rapidly expanding livestock systems are critical upstream measures. Additionally, post-harvest sanitation and cold chain investments for fresh produce in informal markets are essential. Projection models suggest a significant decline in the global burden by 2050 if current trends continue; however, existing paths are still inadequate to reach health equity, emphasizing the urgent need for accelerated food safety capacity-building in low-resource settings. Future validation requires long-term microbiological surveillance, integration of climate-pathogen ecology, and assessment of antimicrobial resistance impacts to support the WHO Global Food Safety Strategy and related Sustainable Development Goals.

CRediT authorship contribution statement

Ziwen Zhou: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Conceptualization. **Yangtai Liu:** Supervision, Investigation, Conceptualization. **Shenghao Yu:** Validation, Software, Methodology. **Haoxin An:** Supervision, Conceptualization. **Yuxin Fu:** Supervision, Conceptualization. **Fan Yang:** Supervision, Conceptualization. **Qingli Dong:** Writing – review & editing, Validation, Resources, Methodology, Conceptualization. **Shaojie Peng:** Writing – review & editing, Validation, Software, Resources, Methodology, Conceptualization.

Declaration of competing interest

No potential conflict of interest was reported by the authors.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijfoodmicro.2026.111883.

Data availability

Data will be made available on request.

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